Cytokines and immunotherapy

Dr. Feyan M. Abdullah
Hawler Medical University
College of Medicine
Microbiology, Physiology and Genetics Dep.
Medical Microbiology Unit

SGL8

Cytokines

- **Cytokines** are **regulatory** soluble proteins or glycoproteins with low-molecular weight orchestrating both innate immunity and the adaptive responses.
- Secreted by immune cells (eg. lymphocytes, macrophages), and a variety of other non immune cells (eg. epithelial and endothelial cells) in the body in response to a number of stimuli.
- Act as messengers that allow the cells of the immune system to communicate with one another to generate a coordinated, robust, **but self-limited response to a target Ag.**
- When secreted in **excess**, or when there exists **an imbalance between proinflammatory and anti-inflammatory cytokines**, resulting in diseases ranging from sepsis to organ transplantation rejection to AIDs occur.

- There are at least 100 different cytokines and growth factors, and are products of one to several genes.
- Cytokines levels can be evaluate in many biological fluids such as serum, blood, stool, saliva, and sweat, provides valuable information regarding the diagnosis, stage, and prognosis of various diseases.

Terms used for cytokines

- Lymphokines, monokines, interleukins, and chemokines
- Some cytokines are known by common names, including:

✓ Interferons (IFNs)
✓ Tumor necrosis factors (TNFs).
✓ Colony stimulating factors (CSFs).
✓ Transforming growth factors (TGFs).

- Cytokines can serve as **biomarkers** of diseases.

Example

The proinflammatory cytokines TNF-α, IL-1, and IL-6 can be detected in the sera of patients with different inflammatory disease like septic shock.

- Recombinant cytokines are **key therapeutic agents**.

Example

- IFN-β for multiple sclerosis.
- IFN-γ for chronic granulomas disease.

- Cytokines can be **targets of therapy**.

Example

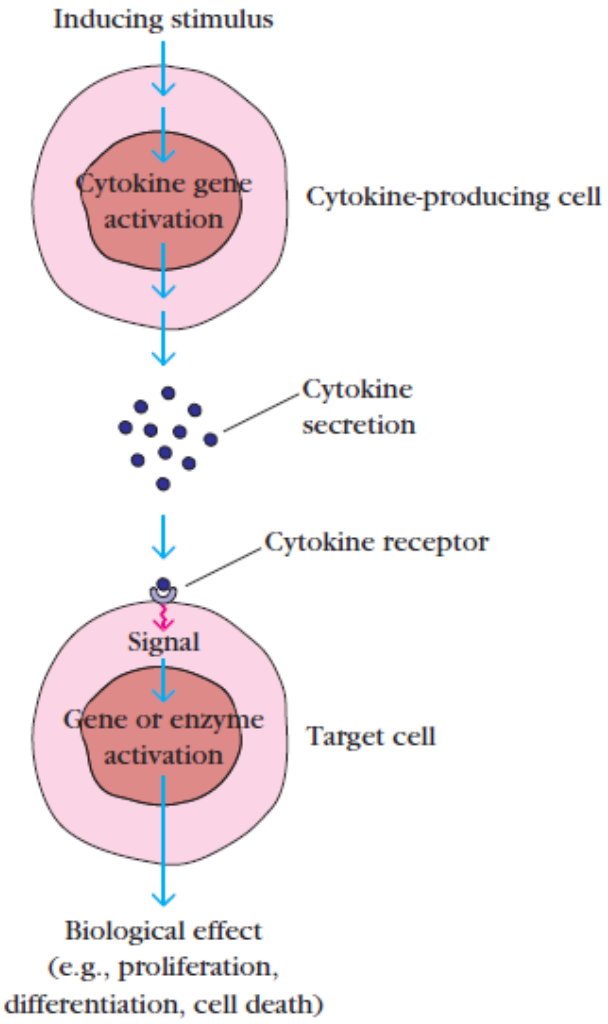
The inhibitors of TNF-α used to manage RA. Inhibitor of IL-2 used in transplantation and cancer.

Properties of Cytokines

- The susceptibility of the target cell to a particular cytokine is determined by the presence of **specific membrane receptors.**

- Cytokines bind to specific receptors on the membrane of target cells, triggering signal-transduction pathways **(action and response)**

- Cytokines and their receptors exhibit **very high affinity** for each other, Because their affinities are so high, cytokines can mediate biological effects at **very small concentrations.**



Cytokine Receptors

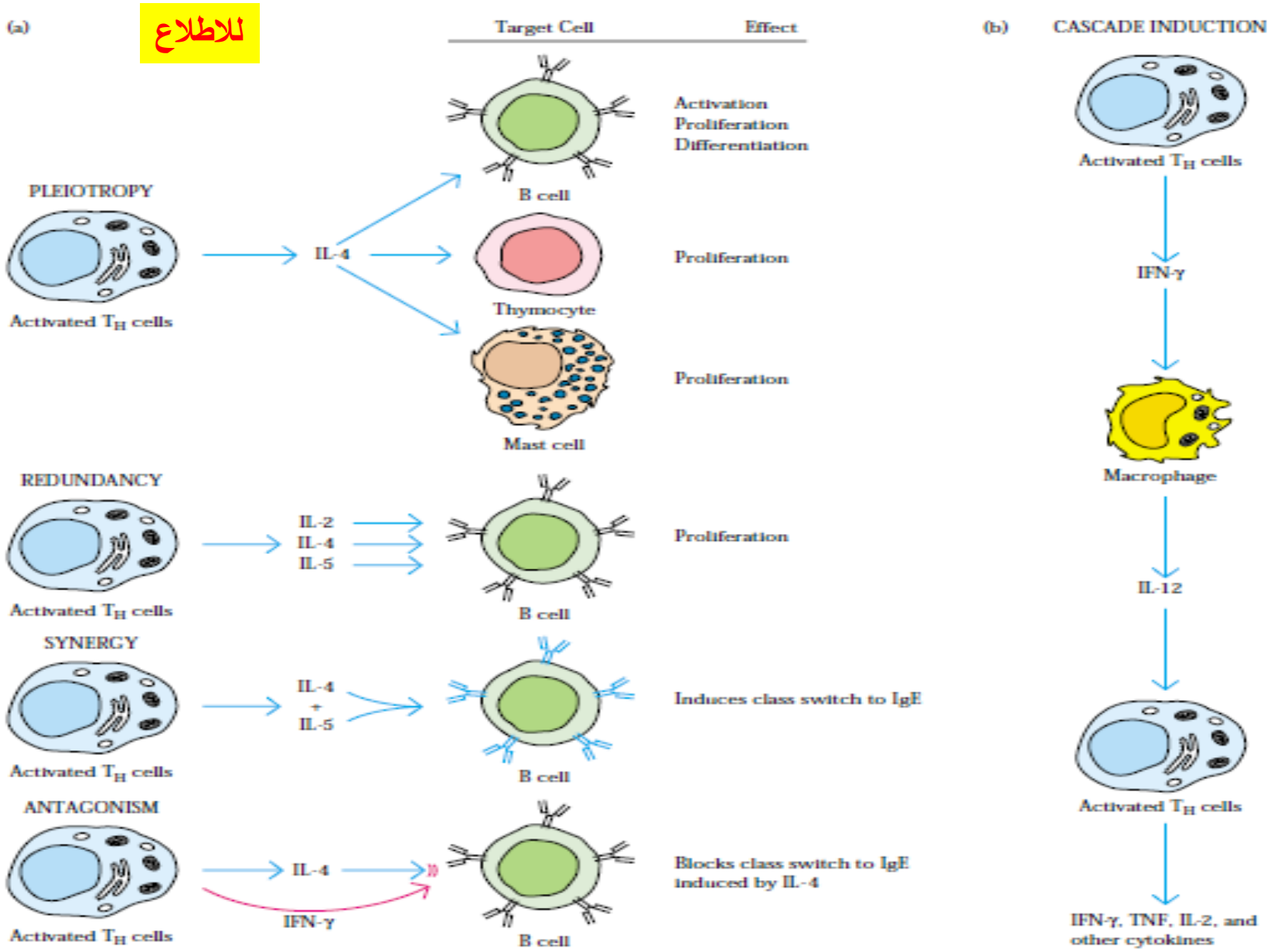
- A given cytokine that has different biological effects on different target cells has a **pleiotropic action**.

- Two or more cytokines that mediate similar functions are said to be **redundant.**

- **Cytokine synergism** occurs when the combined effect of two cytokines on cellular activity is greater than the additive effects of the individual cytokines.

- In some cases, cytokines exhibit **antagonism**; that is, the effects of one cytokine inhibit or offset the effects of another cytokine.

- **Cascade induction** occurs when the action of one cytokine on a target cell induces that cell to produce one or more other cytokines, which in turn may induce other target cells to produce other cytokines.



Cytokine attributes of (a) pleiotropy, redundancy, synergy (synergism), antagonism, and (b) cascade induction.

Biological functions of Cytokines

- Development of cellular and humoral immune responses, induction of the inflammatory response, regulation of hematopoiesis.
- Cytokines can cause changes in the expression of adhesion molecules and chemokine receptors on the target membrane, thus allowing the cell to move from one location to another.
- Cytokines can also signal an immune cell to increase or decrease the activity of particular enzymes or to change its transcriptional program, thereby activating it to proliferate and differentiate.
- Cytokines modulate cell effector functions.
- Cytokines can instruct a cell when to survive and when to die.

Cytokine-Related Diseases

Defects in the complex regulatory networks governing the expression of cytokines and cytokine receptors have been implicated in a number of diseases.

Examples:-

- People with a defective receptor for IFN-Y are susceptible to mycobacterial infections.

- **Cytokine storm** a life-threatening systemic inflammatory syndrome characterized by hyperactivation of immune cells and elevated levels of circulating proinflammatory cytokines for example in case of septic shock in response to microbial infection (eg. *Staphyloccocus aureus*).

- **Cytokine storms** may lead to shock, multi-organ failure, or even death, thus contributing to pathogenesis.

Interleukin-6 (IL-6)

- Has a pleiotropic effect on inflammation, immune response, and hematopoiesis.
- Induces acute phase reactants (eg. CRP…etc), fever, and B-cell proliferation and differentiation.
- IL-6 promotes specific differentiation of naïve $CD4^+$ T cells.
- Involved in the regulation of serum iron and zinc levels via control of their transporters responsible for hypoferremia, anemia and hypozincemia associated with chronic inflammation.
- IL-6 is elevated in the adipose tissue and plasma of obese subjects.
- IL-6 also directly stimulates many cancer cells by activating various cellular signaling pathways that drive cell cycle progression and proliferation.
- IL-6 involve in chemo-resistance.

❖ Immunotherapy:-

Neutralization of IL-6 activity by the IL-6R-specific monoclonal Ab tocilizumab (Actemra) in the treatment of AIDs (eg. arthritis) has been approved.





Tumor necrosis factor alpha (TNF-α)

- ✓ It is a potent **inflammatory mediator** rapidly released after trauma, and infection.
- ✓ It activates **the respiratory burst within neutrophils**, thereby enhancing the killing power of these phagocytes.
- ✓ TNF-α has a **wide variety of metabolic effects**, like decreased insulin sensitivity and increased hepatic production of lipids.
- ✓ TNF-α is Interferes with catabolism in muscle and fat **(resulting in cachexia)** so known as cachectin, cachexin.
- ✓ TNF-α induce acute phase response and other cytokines.
- ✓ TNF-α is an endogenous pyrogens.

Immunotherapy:-

- ✓ Anti-TNF therapeutics such as Infliximab which is a TNF-α blocker in a range of human diseases including Crohn's disease and RA.





Interferon- gamma (IFN-γ)

- IFN-γ is a pleiotropic cytokine with antiviral, antitumor, and immunomodulatory functions.
- IFN-γ increase the expression of class I and II of MHC, increases Ag presentation.
- Enhance Th1 differentiation, and inhibits proliferation of Th2 cells.
- Inhibits allergic reactions (↓IgE production).

Immunotherapy:-

- Actimmune (interferon gamma 1 b) is an interferon gamma used to prevent infections in people with chronic granulomatous disease.



Interleukin-10 (IL-10)

- Inhibition of cytokine production.

- Is an anti-inflammatory cytokine that maintains the balance of the immune response, allowing the clearance of infection while minimizing damage to the host (regulatory cytokines).

- High levels of IL-10 are also produced under conditions that induce anergy and tolerance for example in the development of organ transplant tolerance (enhancement of T regulatory cell development and function).

- IL-10 can also reduce the harmful immune responses elicited in autoimmunity and allergy.

Immunotherapy

Immunotherapy

- **Immunotherapy** is a type of biological therapy (biotherapy) use materials made by the body or in a laboratory to boost, target, or restore a person's immune system.

- It involves the stimulation, enhancement, suppression, or de-sensitization of the immune system by targeting disease-causing virulence factors and improving immunological memory.

- It uses to **fight diseases** such as an autoimmune disorders, allergy, infectious diseases or cancer.

Types of immunotherapy

- **Active immunotherapy**
- **Passive immunotherapy**

Specific active immunotherapy:- Stimulating the host's immune system to generate **Abs** and/or **immune cells** specifically against the target Ag, for example to destroy a tumor or infectious microorganism such as **vaccine therapy.**

- ✓ **Cancer vaccine:-** May contain cancer cell, purified tumor specific Ag …. Example cell based type:-patients cancer cell is cultured with patients own immune system cells and drived back to the same patient….. Dendritic cell (DC) vaccine.

- ✓ **DCs** are professional APCs. Administration of tumor Ag–loaded DCs represents an attractive strategy to generate tumor-specific effector cells.

- ✓ The first **DC vaccine** introduced was for prostate cancer.

Non specific active immunotherapy:- Aims to enhance the overall host immune response.

- ✓ Activating a wide range of immune cells.
- ✓ Many cytokines have the potential to induce non-specific inflammatory responses

- **IFN-α** for treatment of malignant melanoma, in combination with chemotherapy, It is also used to treat certain lymphomas and leukemia.

- **IL-2** for treatment of melanoma and renal cell carcinoma, also IL-2 used in HIV infection where IL-2 production is defective.



Passive immunotherapy

- ✓ Immune molecules are **given to patients** who do not produce them on their own.
- ✓ It do not stimulate a patient immune system to actively respond to a disease in such a way that do vaccines.
- ✓ It involve the transfer of either **Abs** or **cells** into a host.
- ✓ It involve adoptive immune therapy, immunomodulators, and oncolytic virus therapy.

Antibodies (monoclonal Abs)

Useful in fighting diseases because they are specifically designated to only a specific target Ag.

- **Anti-CD20, Rituxan (rituximab)** approved for the treatment of chronic lymphocytic leukemia (CLL) and non-Hodgkin lymphoma.

- **Herceptin (trastuzumab)**, Ab specific for the **oncogene product HER2/Neu** is an approved treatment for breast cancer patients.

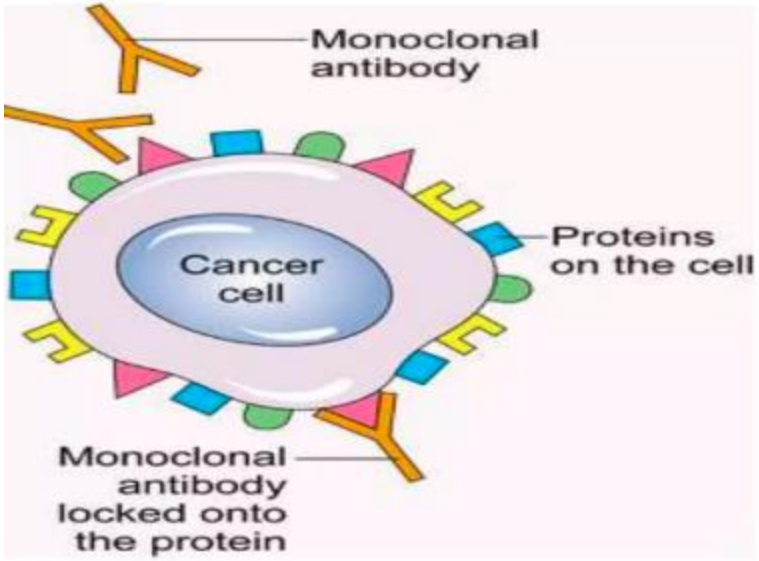
- An **IL-17 blocking Ab** to prevent IL-17 mediated pathogenesis, such as in psoriasis.





- **Omalizumab**, a mAb against IgE, inhibits the binding of IgE to mast cells and basophils. It is effective in moderate and severe allergic asthma and rhinitis.





Immunomodulators:- Regulate or modulate the activity of immune system.

- **Immune checkpoint inhibitors**

- Are a new type of immunotherapy drug used to treat cancer.
- Work by blocking immune checkpoints—the “brakes” of the immune system—that tumors frequently manipulate in order to shut down immune responses and protect themselves.
- As a result, checkpoint inhibitors are able to unleash new immune responses against cancer as well as enhance existing responses to promote elimination of cancer cells.
- The two major immune checkpoint pathways targeted in anti-cancer therapy are: the cytotoxic T-lymphocyte-associated antigen 4 (CTLA-4) and programmed death 1 (PD-1) immune checkpoint pathways.

1. Anti-CTLA-4 – Ipilimumab

Target: CTLA-4 on T cells.

Effect: removes inhibitory signals during T cell activation.

2. Anti-PD-1 – Pembrolizumab, Nivolumab

Target: PD-1 on T cells.

Effect: Restores T-cell activity against tumor cells.

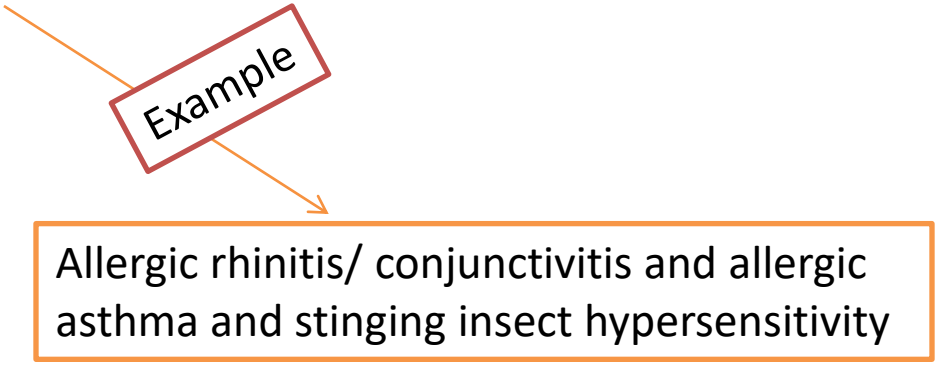
Oncolytic virus (OVs) therapy

- OVs preferentially **target** and kill cancer cells while having minimal to no detrimental effects on normal cells.
- OVs are classified into two categories: naturally occurring and genetically modified.
- OVs mediate tumor cell destruction by **two main mechanisms**: direct lysis of infected cells (oncolysis) and indirect augmentation of host antitumor immunity.
- The recognition and killing of tumor cells are primarily mediated by **natural killer cells** and tumor **antigen-specific CD8+ cytotoxic T lymphocytes**.
- Talimogene laherparepvec (T-VEC), is a genetically modified herpes simplex virus, type 1, approved for the treatment of metastatic melanoma.

enhancing its oncolytic effectiveness

Allergen-specific immunotherapy (AIT)

- Is an immune tolerance–inducing treatment, effective in reducing symptoms of atopic with IgE-mediated allergic.



- Applied to patients who have shown an inadequate response to anti-allergy drugs or who have experienced unacceptable drug side effects.

- Involves the:-

  - subcutaneous administration of gradually increasing quantities of the patient's relevant allergens until a dose is reached that is effective in inducing immunologic tolerance to the allergens.

- **Sublingual tablet** involve regular self-administration of allergen extract under the tongue such odactra.





The primary objectives of allergen-specific immunotherapy are to decrease the symptoms triggered by allergens and to prevent recurrence of the disease in the long-term.

Mechanisms of allergen-specific immunotherapy

- Shift from Th2 immune responses, which are associated with the development of atopic conditions, to Th1 immune responses.

- Production of T regulatory cells that produce the anti-inflammatory cytokine like interleukin 10 (IL-10).

- IL-10 has been shown to reduce levels of IgE, and increase levels of IgG (blocking).
-
- Reduce the release of pro-inflammatory cytokines from mast cells, eosinophils and T cells.

- Decrease the recruitment of mast cells, basophils, and eosinophils to the skin, nose, eye, and bronchial mucosa after exposure to allergens, and reduce the release of mediators, such as histamine, from basophils and mast cells.

Some advantages and disadvantages of immunotherapy

Advantages

- The treatment effect of tumor is good, and the long-term survival rate is significantly improved.
- Wide adaptability. The treatment can control and kill multiple types of tumors.
- The side effects are less than the traditional treatment.

Disadvantages

- There are limitations in the treatment objects and high selectivity for the users.
- The use of immunocheckpoint inhibitors can produce negative regulation, leading to autoimmune diseases.
- Like other treatments, immunotherapy can has side effects from skin rash to more severe one like.
- Treatment costs are high.

References

- Owen JA. et al. Kuby Immunology.7th edition.2009.
- Chapel H, Haeney M, Misbah S, Snowden N. Essentials of Clinical Immunology. 6th ed. 2014.
- Zabriskie JZ. Essential Clinical Immunology. 2009.
- Rabson A., Roitt AM, Delves PJ. (2005).Really Essential Medical Immunology
- Spickett G. Oxford Handbook of Clinical Immunology and Allergy.fourth edition.2020.
- Stevens CD, and Miller LE.,.Clinical Immunology and Serology. A Laboratory Perspective.Fourth Edition.2017.
- Internet sources.